SOFTWARE REQUIREMENTS SPECIFICATION

For

ATTENDLY: Student Event Attendance and Sanctions System

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1. INTRODUCTION

ATTENDLY: Student Event Attendance and Sanctions System is a web-based information system designed to automate student attendance tracking during institutional events through fingerprint biometric verification. The system supports the Supreme Student Council (SSC) in managing events, monitoring student attendance, identifying absences from mandatory events, and administering attendance-related sanctions.

The system will initially be piloted among Bachelor of Science in Information Technology (BSIT) students of JHCSC Pagadian Annex Campus. The initial implementation is intended to establish and evaluate the effectiveness of the system before potential expansion to other academic programs.

1.1 PURPOSE

The purpose of ATTENDLY is to provide an accurate, efficient, and centralized method of recording and monitoring student attendance during institutional events using fingerprint biometric verification.

The system is intended to reduce the limitations of manual attendance recording by automatically capturing attendance information, including the date and time of check-in. It will also monitor absences from mandatory events, determine student attendance standing based on configurable thresholds, and assist SSC officers in reviewing and recording appropriate sanction actions.

1.2 INTENDED AUDIENCE

This document is intended for the individuals and stakeholders involved in the development, use, testing, and evaluation of ATTENDLY. Students will use the system to view event schedules, check in to events, and monitor their attendance records and standing. SSC Officers will manage events, attendance, and reports within their own scope, and recommend sanctions for Administrator review. Administrators will manage student accounts, SSC Officer

accounts, fingerprint enrollment, and system-wide reports. System Developers and Testers will use this document as a guide for system development and testing.

1.3 PRODUCT SCOPE

ATTENDLY addresses the challenges of manually recording and monitoring student attendance during institutional events, including inaccurate records, delayed monitoring, and difficulty tracking mandatory-event absences and sanctions. The system aims to provide an accurate and efficient attendance management process through fingerprint biometric verification, centralized attendance records, and automated sanction monitoring.

The system is intended for JHCSC students and SSC officers. Students will use the system to check in to events and view their attendance information, while SSC officers will manage events and attendance records within their own scope, notifications, and recommend sanctions for Administrator review. Administrators will manage student accounts, SSC Officer accounts, fingerprint enrollment, and system-wide reports.

The initial implementation will be limited to BSIT students of JHCSC Pagadian Annex Campus**.** It will exclude other academic programs, integration with the school's grading or registrar system, non-fingerprint biometric technologies, payment processing, and parent or guardian access. Future enhancements may include expansion to other academic programs, multi-campus deployment, mobile application support, and integration with other school systems.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

- SSC (Supreme Student Council) **-** is a group of student leaders who represent and serve the students in the school.
- Biometric - Technology used to identify a person through unique physical characteristics.

- Fingerprint Template - Digitally stored representation of a fingerprint used for verification.
- Mandatory Event - An event that students are required to attend.
- Optional Event - An event that students may attend but does not affect sanctions.
- Sanction - An action or penalty given for violating attendance policies.
- ATTENDLY - The Student Event Attendance and Sanctions System developed for managing student event attendance and sanctions.
- Fingerprint Enrollment - The process of registering a student's fingerprint template in the system for future verification.
- Attendance Record - A stored record indicating a student's attendance or absence from an event.
- Fingerprint Verification - The process of comparing a captured fingerprint against a registered fingerprint template to verify a student's identity.

1.5 DOCUMENT CONVENTIONS

This section defines the labeling, formatting, and documentation conventions used throughout this SRS to ensure consistency and clarity.

- Requirement Identification - Each requirement in this document is assigned a unique identifier consisting of a prefix and a three-digit sequence number: FR-XXX for Functional Requirements, NFR-XXX for Non-Functional Requirements, BR-XXX for Business Rules, and TC-XXX (or TC-NFR-XXX for non-functional test cases) for Test Cases.
- Priority Levels - Each functional requirement is assigned a priority of High, Medium, or Low, as defined in Section 4.3.1, to indicate its relative importance to system development.
- Formatting Conventions - Statements containing “shall” indicate mandatory requirements. Defined terms such as Mandatory Event and Optional Event are

capitalized when referenced. Section and sub-section headings follow a numbered outline format (e.g., 1.1, 1.2), and structured requirement data is presented in tables for clarity.

Acceptance Criteria Format - Each functional requirement in Section 4.1 is paired with an acceptance criteria entry presented in a five-column format — Input, Process, Output, Expected System Behavior, and Acceptance Criteria — describing the condition under which the requirement is considered satisfied.

1.6 REFERENCES

The following sources were reviewed and used as a basis for this SRS, including related studies on biometric and fingerprint-based attendance systems.

Adedoyin, M. A., Shoewu, O. O., Yussuff, A. I. O., Adenowo, A. A., Shitta, A. H., & Okedokun, O. (2024). Development of an IoT-Based Biometric Attendance Management System. *FUOYE Journal of Engineering and Technology*.

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2. PROJECT PROPOSAL AND TEAM FORMATION

2.1 APPROVED PROJECT CONCEPT

ATTENDLY: Student Event Attendance and Sanctions System is a proposed web-based system designed to automate student attendance monitoring during institutional events through fingerprint biometric verification. The project was selected to address the limitations of manual attendance recording, particularly inaccurate records, delayed monitoring, and difficulty in tracking mandatory-event absences and attendance-related sanctions. The system aims to provide a centralized and efficient solution for managing attendance and sanctions.

2.2 TEAM FORMATION

The project will be developed by a two-member team, with responsibilities divided according to the major activities required for system development. Both members will collaborate throughout the planning, design, development, testing, documentation, and presentation of ATTENDLY.

Team Member	Role	Responsibilities
Kirt M. Dela Torre	Project Leader / System Developer / Database Designer	Leads and coordinates the project, oversees system development, designs and manages the database, develops core system functions, assists with system integration, participates in testing, and leads the project presentation.
Nicole P. Jansol	System Analyst / UI Designer / Documentation Lead	Analyzes and documents system requirements, designs and develops the user interface, prepares project documentation, assists in system development and testing, and participates in project presentation.

2.3 PROBLEM STATEMENT

The project addresses the challenges associated with manual attendance monitoring during institutional events. Traditional methods may result in inaccurate or incomplete attendance records, delayed monitoring, difficulty identifying students who are absent from mandatory events, and inefficient tracking of sanctions. ATTENDLY aims to address these issues by providing automated fingerprint-based attendance recording and centralized monitoring of attendance and sanction information.

2.4 PROJECT OBJECTIVES

The project aims to develop a system that:

- Automates student attendance recording during institutional events.
- Uses fingerprint biometric verification for student identification.
- Centralizes event, attendance, and sanction records.
- Allows SSC officers to efficiently manage events and attendance.
- Improves the accuracy and accessibility of attendance information.
- Supports systematic monitoring and recording of sanctions.
- Generates attendance and sanction reports for authorized users.
- Provides notifications related to attendance and sanctions.
- Protects system information through role-based access control.

2.5 TARGET USERS

The primary users of ATTENDLY are BSIT students, SSC officers, and Administrators. Students will use the system to check in to institutional events through fingerprint verification and view their attendance records and current standing. SSC officers will serve as event organizers within their own scope, managing their own events and attendance records, and

recommending sanction actions for Administrator review. Administrators will serve as the authorized **administrators** of the system, managing student accounts, SSC Officer accounts, fingerprint enrollment, and system-wide reports.

2.6 CORE FEATURES

The proposed system will include user and account management, authentication and role-based access, fingerprint registration and verification, biometric attendance recording, event management, attendance monitoring and reporting, sanction management, notifications, and centralized data storage.

2.7 SCOPE BOUNDARIES

The initial implementation of ATTENDLY will focus on BSIT students of JHCSC Pagadian Annex Campus. The system will cover institutional event attendance and attendance-related sanctions. It will exclude online payment processing, academic grading, payroll or employee management, multi-campus deployment, a dedicated mobile application, and advanced analytics. These features may be considered for future enhancement based on institutional requirements, available resources, and project evaluation.

2.8 COMPARISON WITH SIMILAR OR EXISTING SYSTEMS

Existing attendance systems use various methods such as manual attendance sheets, QR codes, RFID, and biometric technologies. Manual systems are simple but may require more time and can make attendance monitoring and record management difficult. QR-based systems provide faster recording but may allow attendance codes to be shared. RFID systems can automate identification but require compatible RFID equipment. Biometric systems provide identity-based verification but require specialized hardware and proper management of biometric data.

ATTENDLY is a web-based attendance management system specifically designed for institutional events and attendance-related sanctions. It combines fingerprint verification with

event management, centralized attendance records, automatic absence monitoring, sanction tracking, notifications, and reporting. Unlike basic attendance systems that primarily focus on recording check-ins, ATTENDLY provides a centralized web-based platform for both attendance monitoring and sanction management.

2.9 PROPOSAL APPROVAL AND SUPPORTING DOCUMENTS

The project proposal, including the approved concept, objectives, scope, and system features, shall be supported by the team's project proposal document, adviser feedback, presentation materials, and approval evidence, where applicable. These documents shall serve as references for confirming that the development of ATTENDLY follows the approved project direction and requirements.

3. OVERALL DESCRIPTION

3.1 PRODUCT PERSPECTIVE

ATTENDLY is a new web-based student event attendance management system developed to replace the manual process of recording and monitoring attendance during institutional events. The system will integrate fingerprint biometric verification with event management, attendance monitoring, sanction management, notifications, and reporting. It will operate as a standalone system during its initial implementation and will not be integrated with the school's existing grading or registrar systems.

3.2 USER CLASSES AND CHARACTERISTICS

ATTENDLY will serve three primary user groups: BSIT Students, SSC Officers, and Administrators. BSIT Students will use the system to check in to events through fingerprint verification, view upcoming events, access their attendance history, and monitor their attendance standing. SSC Officers will serve as event organizers within their own scope, managing their own events and attendance records, monitoring attendance, and recommending sanction actions for Administrator review. Administrators will serve as the authorized administrators of the system and will manage student accounts, SSC Officer accounts, fingerprint enrollment, sanction finalization, and system-wide reports. All user groups are expected to have basic computer and web browsing skills, while SSC Officers and Administrators should have additional knowledge of event and attendance management.

3.3 OPERATING ENVIRONMENTS

ATTENDLY shall operate in a web-based environment using the following:

- Devices: Desktop computers and laptops
- Operating Systems: Windows 10 or later, or other operating systems capable of running supported web browsers
- Web Browsers: Google Chrome, Microsoft Edge, or Mozilla Firefox
- Network: Stable local network or Internet connection

- Database: A relational database management system such as MySQL
- Server: Web server connected to the centralized database
- Biometric Device: Compatible fingerprint scanner

3.4 DESIGN AND IMPLEMENTATION CONSTRAINTS

The development of ATTENDLY is subject to hardware, software, security, and data privacy constraints. As a **web-based system**, it requires a supported web browser and a stable internet or school network connection to access the system. The system must also be compatible with the fingerprint scanner and its required SDK or driver for biometric verification. Offline attendance processing will not be supported in the initial version. Access to system functions shall be controlled through role-based access, while personal and biometric information shall be protected using appropriate security measures. The system shall comply with applicable data privacy requirements, particularly the Data Privacy Act of 2012 (Republic Act No. 10173).

3.5 ASSUMPTIONS AND DEPENDENCIES

The system is assumed to be web-based and initially used on a single campus. SSC officers will serve as the event organizers, so no separate event organizer role is required. The system also depends on a compatible fingerprint device, stable network connection, and database server.

4. REQUIREMENTS ANALYSIS

4.1 FUNCTIONAL REQUIREMENTS

ID	Functional Requirement	Source	Priority	Module
FR-001	The system shall allow Administrators to create, edit, and deactivate student accounts system-wide.	Administrator	High	Administration
FR-002	The system shall authenticate Students and SSC Officers using valid login credentials.	Student/SSC Officer	High	Authentication
FR-003	The system shall allow Administrators to enroll and re-enroll a student's fingerprint template.	Administrator	High	Administration
FR-004	The system shall verify a student's fingerprint during event check-in.	Student	High	Biometric Attendance
FR-005	The system shall record attendance after successful fingerprint verification.	SCC Officer	High	Attendance Management
FR-006	The system shall prevent duplicate attendance records for the same student and event.	Student/SSC Officer	High	Attendance Management

FR-007	The system shall allow SSC Officers to create, edit, and cancel institutional events.	SSC Officer	High	Event Management
FR-008	The system shall allow SSC Officers to classify events as mandatory or optional.	SSC Officer	High	Event Management
FR-009	The system shall allow SSC Officers to monitor attendance and identify students absent from mandatory events.	SSC Officer	High	Attendance Monitoring
FR-010	The system shall automatically count mandatory-event absences and determine student standing based on configured thresholds.	SSC Officer	High	Sanction Management
FR-011	The system shall allow SSC Officers to flag a student for a sanction, with a reason and severity, for Administrator review when the applicable threshold is reached.	SSC Officer	High	Sanction Management
FR-012	The system shall allow students to view attendance records and standing, and	Student/SSC Officer	Medium	Dashboard & Reporting

	SSC Officers to generate attendance and sanction reports for their own events.			
FR-013	The system shall allow Administrators to create, approve, edit, and deactivate SSC Officer accounts.	Administrator	High	Administration
FR-014	The system shall allow Administrators to audit fingerprint enrollment activity, including which Administrator enrolled or re-enrolled each record and when.	Administrator	Medium	Administration
FR-015	The system shall allow Administrators to finalize or override a recommended sanction into an official Warning or Probation status.	Administrator	High	Administration
FR-016	The system shall allow Administrators to generate system-wide attendance and sanction reports across all SSC Officers and events.	Administrator	Medium	Administration

FR-017	The system shall allow SSC Officers to view student records relevant to their own scope, without the ability to create, edit, or delete accounts.	SSC Officer	Medium	User Management

ID	Input	Process	Output	Expected System Behavior	Acceptance Criteria
FR-001	Student information and Administrator credentials	Validate and save, edit, or deactivate account information	Student account	The system creates, updates, or deactivates an account when a valid request is submitted by an Administrator.	A valid student account is successfully created, edited, or deactivated by an Administrator.
FR-002	Username/email and password	Validate credentials and user role	User dashboard or error message	The system grants access only to authorized users.	Valid credentials allow login, while invalid credentials are rejected.
FR-003	Student information and fingerprint scan	Capture and enroll or re-enroll fingerprint template	Registered fingerprint template	The system links the enrolled fingerprint template to the correct student account when	A student's fingerprint is successfully enrolled or re-enrolled by an Administrator

				submitted by an Administrator.	and associated with the correct account.
FR-004	Fingerprint scan and event	Match scanned fingerprint with registered template	Verification result	The system identifies the student when the fingerprint matches.	A valid fingerprint successfully identifies the correct student.
FR-005	Verified student and event	Create and save attendance record with date and time	Attendance record	The system automatically records attendance after successful verification.	A timestamped attendance record is stored for the correct student and event.
FR-006	Student ID and event ID	Check existing attendance record	Confirmation or duplicate warning	The system prevents multiple attendance records for the same event.	A second attendance attempt for the same event is rejected.
FR-007	Event name, date, time, venue, and event type	Validate and save event information	Event record	The system reflects authorized event changes.	An event can be successfully created, edited, or cancelled.
FR-008	Event classification	Save event type	Mandatory/ optional status	The system applies the selected	The correct classification is

				classification to the event.	displayed and stored.
FR-009	Student and attendance records	Compare expected attendance with recorded attendance	Attendance and absence records	The system identifies students who do not have valid attendance for mandatory events.	Absent students are correctly identified.
FR-010	Attendance records and thresholds	Calculate absences and compare them with thresholds	Absence count and standing	The system automatically updates the student's standing.	The correct absence count and standing are displayed.
FR-011	Student information, sanction reason, and severity	Validate and save a recommended sanction for review	Recommended sanction record	The system records the recommended sanction and routes it to an Administrator for review.	A valid sanction recommendation is successfully recorded and routed for Administrator review.
FR-012	Attendance and sanction data	Retrieve and organize records within the user's assigned scope	Dashboard information and reports	Users can access information according to their assigned role and scope.	Students can view their own records, while SSC Officers can generate reports limited

					to their own events.
FR-013	SSC Officer information	Validate and save, approve, edit, or deactivate officer account information	SSC Officer account	The system creates, approves, edits, or deactivates an SSC Officer account only when submitted by an Administrator.	A valid SSC Officer account is successfully created, approved, edited, or deactivated.
FR-014	Fingerprint enrollment records	Retrieve and display enrollment activity logs	Enrollment audit report	The system displays who enrolled or re-enrolled each fingerprint record and when.	Administrators can successfully view a complete and accurate enrollment audit trail.
FR-015	Recommended sanction record and Administrator decision	Validate and finalize or override the recommended sanction	Official sanction record (Warning/Probation)	The system updates the sanction status only when finalized or overridden by an Administrator.	A recommended sanction is successfully converted into an official Warning or Probation status.

FR-016	Attendance and sanction data across all events	Retrieve and organize system-wide records	System-wide report	The system compiles attendance and sanction data across all SSC Officers and events for Administrators.	Administrators can successfully generate accurate cross-event, system-wide reports.
FR-017	Student records within the SSC Officer's scope	Retrieve and display read-only student information	Scoped student list	The system displays only student records relevant to the SSC Officer's own scope, without edit access.	SSC Officers can view relevant student records but cannot create, edit, or delete them.

4.2 NON-FUNCTIONAL REQUIREMENTS

ID	Category	Requirement	Acceptance Threshold / Standard
NFR-001	Performance	The system shall verify a fingerprint within 3 seconds under normal operating conditions.	At least 95% of verification attempts shall be completed within 3 seconds during testing.
NFR-002	Security	The system shall restrict system functions according to the user's assigned role.	Students shall not be able to access SSC Officer functions.

NFR-003	Data Privacy	The system shall protect personal and fingerprint information in accordance with the Data Privacy Act of 2012 (RA 10173).	Unauthorized users shall not be able to access protected information.
NFR-004	Reliability	The system shall maintain at least 99% availability during scheduled event hours, subject to network and server availability.	The system shall meet the 99% availability target during the defined testing period.
NFR-005	Usability	The system shall provide clear navigation, readable information, and understandable system messages.	Representative users shall be able to perform core tasks after a short orientation.
NFR-006	Compatibility	The web-based system shall function correctly on supported modern web browsers.	Core functions shall operate on Google Chrome, Microsoft Edge, and Brave.

4.3 REQUIREMENTS PRIORIZATION AND TRACEABILITY

4.3.1 REQUIREMENT PRIORITIZATION

Priority	Description
High	Essential requirements necessary for the core operation of the system.
Medium	Important requirements that support system efficiency, monitoring, and reporting.
Low	Optional requirements that may be considered for future enhancement.

4.3.2 REQUIREMENT TRACEABILITY MATRIX

The traceability matrix ensures that each approved requirement is addressed during system design, development, and testing.

ID	Source	Objective	Module
FR-001	Administrator	Account Management	Administration
FR-002	Student/SSC Officer	Secure Access	Authentication
FR-003	Administrator	Fingerprint Registration	Administration
FR-004	Student	Biometric Attendance	Attendance
FR-005	Student	Attendance Recording	Attendance
FR-006	Student/SSC Officer	Data Integrity	Attendance
FR-007	SSC Officer	Event Management	Event Management
FR-008	SSC Officer	Event Classification	Event Management
FR-009	SSC Officer	Attendance Monitoring	Attendance Monitoring
FR-010	SSC Officer	Sanction Monitoring	Sanction Management
FR-011	SSC Officer	Sanction Management	Sanction Management
FR-012	Student/SSC Officer	Reporting	Dashboard & Reporting
FR-013	Administrator	Officer Account Management	Administration
FR-014	Administrator	Enrollment Audit	Administration
FR-015	Administrator	Sanction Finalization	Administration
FR-016	Administrator	System-Wide Reporting	Administration
FR-017	SSC Officer	Scoped Student Visibility	User Management
NFR-001	System Requirement	Performance	Fingerprint Verification
NFR-002	Security Requirement	Access Control	Authentication
NFR-003	Privacy Requirement	Data Protection	Security

NFR-004	System Requirement	Reliability	System
NFR-005	User Requirement	Usability	User Interface
NFR-006	System Requirement	Compatibility	Web Interface

ID	Related Design Component	Test Case	Status
FR-001	Account Interface	TC-001	Planned
FR-002	Login Interface	TC-002	Planned
FR-003	Fingerprint Interface	TC-003	Planned
FR-004	Fingerprint Check-in Interface	TC-004	Planned
FR-005	Attendance Database	TC-005	Planned
FR-006	Attendance Validation	TC-006	Planned
FR-007	Event Interface	TC-007	Planned
FR-008	Event Form	TC-008	Planned
FR-009	Attendance Dashboard	TC-009	Planned
FR-010	Standing Module	TC-010	Planned
FR-011	Sanction Interface	TC-011	Planned
FR-012	Report Interface	TC-012	Planned
FR-013	Officer Account Interface	TC-013	Planned
FR-014	Enrollment Audit Interface	TC-014	Planned
FR-015	Sanction Finalization Interface	TC-015	Planned
FR-016	System-Wide Report Interface	TC-016	Planned
FR-017	Scoped Student Interface	TC-017	Planned
NFR-001	Verification Process	TC-NFR-001	Planned
NFR-002	Role-Based Access	TC-NFR-002	Planned

NFR-003	Data Protection Controls	TC-NFR-003	Planned
NFR-004	Server/System Availability	TC-NFR-004	Planned
NFR-005	Web Interface	TC-NFR-005	Planned
NFR-006	Browser Compatibility	TC-NFR-006	Planned

4.4 BUSINESS RULES

The following business rules define the policies and conditions governing the operation of the web-based ATTENDLY system.

ID	Business Rules
BR-001	Only authenticated users shall access protected features of the web-based system.
BR-002	Students shall only view their own attendance records, standing, and notifications.
BR-003	Only authorized SSC Officers shall manage events and attendance, and recommend sanctions, within their own scope.
BR-004	A student must have a registered fingerprint before using biometric attendance through the web-based system.
BR-005	Each event shall be classified as **Mandatory** or **Optional**.
BR-006	Only attendance from mandatory events shall be included in absence and sanction monitoring.
BR-007	Attendance shall be recorded only after successful fingerprint verification.
BR-008	The system shall prevent duplicate attendance records for the same student and event.
BR-009	Students who reach the configured absence threshold shall be flagged for sanction review.

BR-010	Only authorized SSC Officers shall record and recommend sanction actions for Administrator review.
BR-011	Cancelled events shall not accept attendance records or count as absences.
BR-012	Students shall not be allowed to modify or delete their attendance or sanction records.
BR-013	Only authorized Administrators shall create, edit, or deactivate student and SSC Officer accounts, and manage fingerprint enrollment.
BR-014	Only authorized Administrators shall finalize or override a recommended sanction into an official Warning or Probation status.

4.5 EXTERNAL INTERFACE REQUIREMENTS

4.5.1 User Interface

ATTENDLY shall provide a web-based user interface accessible through supported web browsers on desktop or laptop computers. The STUDENT INTERFACE shall provide access to login, upcoming events, fingerprint check-in, attendance history, attendance standing, and notifications. The SSC OFFICER INTERFACE shall provide access to event management, attendance monitoring, sanction recommendation, threshold configuration, reporting on their own events, and notifications. The ADMINISTRATOR INTERFACE shall provide access to student and SSC Officer account management, fingerprint enrollment and enrollment audit, sanction finalization, and system-wide reporting. The interface shall provide clear navigation, readable information, input validation, confirmation messages, and appropriate error messages.

4.5.2 Hardware Interface

ATTENDLY shall use a fingerprint biometric device for student identification and attendance verification. The device shall capture and verify registered fingerprints during event

check-in and communicate with the web-based system through its required SDK, driver, or asupported integration method.

4.5.3 Software Interface

ATTENDLY shall operate through supported modern web browsers and communicate with the web server and centralized database. The database shall store student accounts, fingerprint templates, event information, attendance records, sanction records, and notifications.

4.5.4 Communication Interface

ATTENDLY shall use standard web-based communication protocols between the user's browser and the web server. A stable Internet or school network connection shall be required to access the system and process fingerprint-based attendance. HTTPS shall be used to protect information transmitted between the user's browser and the server where supported by the deployment environment